

A Unified Gray Hammer–Rosen Model for Surface- and Bath-Temperature Drives

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Abstract

A compact Hammer–Rosen-type model is developed for planar gray, two-temperature Marshak waves. The model allows independent Rosseland and absorption opacity power laws and can be driven either by a prescribed surface radiation temperature $T_s(t)$ or by a prescribed bath temperature $T_{\text{bath}}(t)$ through a Marshak boundary condition. The radiation profile uses the first-order Hammer–Rosen curvature correction. The material profile is the same radiation shape multiplied by a monotone factor $\exp[-s(t)y]$, where the single scalar $s(t)$ is obtained from an areal material-energy closure. The surface-driven model requires two scalar physical ODEs, three running time quadratures, and one scalar root. The bath-driven model adds only one scalar ODE for $T_s(t)$. No spatial mesh, diffusion matrix, PDE solve, or history-dependent integral equation is required. A fully self-contained implementation recipe is given in the final section.

Contents

1	Gray diffusion problem	1
2	Surface material temperature	2
3	Common-shape gray Hammer–Rosen front	2
3.1	Gray storage coordinate	2
3.2	Base front	3
4	First-order radiation profile and the two boundary drives	4
4.1	Prescribed surface radiation temperature	4
4.2	Prescribed bath temperature	4
5	Monotone material-profile closure	5
6	Profile moments and their closed forms	5
7	Areal material-energy equation	6
8	Final front correction	7

9 Numerical implementation and complexity	8
9.1 Recommended SciPy functions	8
9.2 Surface-temperature drive	8
9.3 Bath-temperature drive	9
10 Numerical illustration	9
11 Accuracy and limitations	12
12 Self-contained summary of the working model	14

1 Gray diffusion problem

Consider a planar, fixed-density material occupying $x > 0$. Hydrodynamic motion is neglected. The radiation energy density and material energy density are

$$E_r = a_{\text{rad}} T_r^4, \quad U = f \rho^{1-\mu} T^\beta, \quad a_{\text{rad}} = \frac{4\sigma_{\text{SB}}}{c}. \quad (1.1)$$

The Rosseland and absorption opacities are independent power laws,

$$\frac{1}{\kappa_R} = g T^\alpha \rho^{-\lambda}, \quad \frac{1}{\kappa_a} = g' T^{\alpha'} \rho^{-\lambda'}. \quad (1.2)$$

Thus (α, λ, g) are the Rosseland parameters and (α', λ', g') are the absorption parameters.

The underlying gray diffusion equations are

$$\frac{\partial}{\partial t} (a_{\text{rad}} T_r^4) = \frac{\partial}{\partial x} \left[\frac{c g}{3} \rho^{-1-\lambda} T^\alpha \frac{\partial}{\partial x} (a_{\text{rad}} T_r^4) \right] - \frac{a_{\text{rad}} c}{g'} \rho^{1+\lambda'} T^{-\alpha'} (T_r^4 - T^4), \quad (1.3)$$

$$\frac{\partial}{\partial t} (f \rho^{1-\mu} T^\beta) = \frac{a_{\text{rad}} c}{g'} \rho^{1+\lambda'} T^{-\alpha'} (T_r^4 - T^4). \quad (1.4)$$

The reduced model derived below does not solve (1.3)–(1.4) pointwise.

Two boundary specifications are considered.

Surface-temperature drive.

$$\boxed{T_r(0, t) = T_s(t)}, \quad (1.5)$$

where $T_s(t)$ is prescribed.

Bath-temperature drive. The surface radiation temperature is unknown and the incoming bath is imposed through the Marshak condition

$$\boxed{F_s(t) = \frac{a_{\text{rad}} c}{2} [T_{\text{bath}}^4(t) - T_s^4(t)], \quad T_s(t) = T_r(0, t)}, \quad (1.6)$$

where $F_s > 0$ denotes flux into the material.

Define the two standard Hammer–Rosen parameters

$$\boxed{\varepsilon = \frac{\beta}{4 + \alpha}, \quad C = \frac{4a_{\text{rad}}cg}{3(4 + \alpha)f}\rho^{-2+\mu-\lambda}.} \quad (1.7)$$

The exponent of the leading front profile will always be written explicitly as $1/(4 + \alpha - \beta)$; no separate symbol is introduced for it.

2 Surface material temperature

Let

$$\boxed{T_{m,0}(t) = T(0, t)} \quad (2.1)$$

be the surface material temperature. Evaluating the local material balance at $x = 0$ gives

$$\boxed{\frac{d}{dt} [f\rho^{1-\mu}T_{m,0}^\beta(t)] = \frac{a_{\text{rad}}c}{g'}\rho^{1+\lambda'}T_{m,0}^{-\alpha'}(t) [T_s^4(t) - T_{m,0}^4(t)].} \quad (2.2)$$

Equivalently,

$$\boxed{\dot{T}_{m,0}(t) = \frac{a_{\text{rad}}c}{\beta fg'}\rho^{\mu+\lambda'}T_{m,0}^{1-\alpha'-\beta}(t) [T_s^4(t) - T_{m,0}^4(t)].} \quad (2.3)$$

This equation naturally gives $T_{m,0} < T_s$ during heating and may give $T_{m,0} > T_s$ during cooling.

Equation (2.2) is one scalar, generally stiff ODE. In Python it should be integrated with `solve_ivp`, using `method="Radau"` or `method="BDF"`. At a perfectly cold start, a code may internally evolve the regular power $T_{m,0}^{\alpha'+\beta}$, but that auxiliary variable is only a numerical device and is not part of the analytical model.

3 Common-shape gray Hammer–Rosen front

3.1 Gray storage coordinate

For the leading common-shape reduction, write locally

$$T_r(x, t) = T_s(t)q(x, t), \quad T(x, t) = T_{m,0}(t)q(x, t), \quad 0 \leq q \leq 1. \quad (3.1)$$

The ratio of radiation energy to material energy at the surface is

$$\boxed{R(t) = \frac{a_{\text{rad}}T_s^4(t)}{f\rho^{1-\mu}T_{m,0}^\beta(t)}.} \quad (3.2)$$

At an arbitrary profile value q , the corresponding ratio is $R(t)q^{4-\beta}$. The gray storage coordinate entering the HR reduction is therefore

$$Z(q; R) = (4 + \alpha - \beta) \int_0^q \frac{\xi^{3+\alpha-\beta}}{1 + R\xi^{4-\beta}} d\xi. \quad (3.3)$$

The surface value $Z(1; R)$ is the gray storage factor

$$\boxed{G(R) = (4 + \alpha - \beta) \int_0^1 \frac{\xi^{3+\alpha-\beta}}{1 + R\xi^{4-\beta}} d\xi.} \quad (3.4)$$

The denominator is the normalized total storage $1 + R\xi^{4-\beta}$; thus G reduces the effective drive whenever radiation energy is non-negligible.

The integral has the exact Gauss-hypergeometric form

$$G(R) = {}_2F_1\left(1, \frac{4+\alpha-\beta}{4-\beta}; 1 + \frac{4+\alpha-\beta}{4-\beta}; -R\right). \quad (3.5)$$

The identity follows from

$${}_2F_1(a, b; c; z) = \frac{\Gamma(c)}{\Gamma(b)\Gamma(c-b)} \int_0^1 u^{b-1} (1-u)^{c-b-1} (1-zu)^{-a} du, \quad (3.6)$$

valid initially for $\Re c > \Re b > 0$ and elsewhere by analytic continuation. In Python, (3.5) is evaluated directly with `hyp2f1`.

3.2 Base front

Define temporarily the running history

$$\mathcal{I}(t) = \int_0^t T_s^4(t') T_{m,0}^\alpha(t') G(R(t')) dt'. \quad (3.7)$$

Applying the first-order HR front moment to the gray storage coordinate gives

$$x_{F,0}^2(t) = \frac{2+\varepsilon}{1-\varepsilon} C T_{m,0}^{-\beta}(t) \mathcal{I}(t). \quad (3.8)$$

Equivalently, without introducing $\mathcal{I}$,

$$x_{F,0}^2(t) = \frac{2+\varepsilon}{1-\varepsilon} C T_{m,0}^{-\beta}(t) \int_0^t T_s^4(t') T_{m,0}^\alpha(t') G(R(t')) dt'. \quad (3.9)$$

Numerically this is one scalar running quadrature. It is advanced together with the ODE states; no past time array needs to be reintegrated at every step.

4 First-order radiation profile and the two boundary drives

The normalized first-order HR radiation profile is

$$T_r(x, t) = T_s(t) \left[\left(1 - \frac{x}{x_F(t)}\right) \left(1 + B(t) \frac{x}{x_F(t)}\right) \right]^{1/(4+\alpha-\beta)}. \quad (4.1)$$

The coefficient $B(t)$ is the first HR curvature correction. The profile requires $B(t) > -1$ so that its base remains nonnegative on the wave interval.

4.1 Prescribed surface radiation temperature

For prescribed $T_s(t)$, the HR correction is

$$B(t) = \frac{\varepsilon}{2} \left[1 - \frac{2 + \varepsilon}{1 - \varepsilon} \frac{\int_0^t T_s^4(t') T_{m,0}^\alpha(t') G(R(t')) dt'}{T_s^4(t) T_{m,0}^\alpha(t) G(R(t))} \times \frac{d}{dt} \ln T_s^{4+\alpha}(t) \right]. \quad (4.2)$$

This is an algebraic post-processing formula once the base-front history is known. For a tabulated drive, use a shape-preserving interpolant such as `scipy.interpolate.PchipInterpolator`; its analytical derivative can be obtained with `PchipInterpolator.derivative()`. If the tabulated pulse contains measurement noise, regularize only this derivative, not the drive used in the energy quadratures.

4.2 Prescribed bath temperature

At the surface, the diffusion flux predicted by (4.1), using the base front inside the boundary closure, is

$$F_s(t) = \frac{4a_{\text{rad}}cg}{3(4 + \alpha - \beta)x_{F,0}(t)} \rho^{-1-\lambda} T_{m,0}^\alpha(t) T_s^4(t) [1 - B(t)]. \quad (4.3)$$

Equating this to the Marshak flux (1.6) gives

$$B(t) = 1 - \frac{3(4 + \alpha - \beta)x_{F,0}(t)\rho^{1+\lambda}}{8gT_{m,0}^\alpha(t)} \left[\frac{T_{\text{bath}}^4(t)}{T_s^4(t)} - 1 \right]. \quad (4.4)$$

Using $x_{F,0}$ rather than the final moment-corrected front keeps the boundary subsystem sequential; replacing it by x_F would introduce a higher-order implicit feedback.

Equating (4.4) to the HR curvature relation gives one scalar ODE for the unknown surface radiation temperature:

$$\dot{T}_s(t) = \frac{2T_s^5(t)T_{m,0}^\alpha(t)G(R(t))}{\varepsilon(4 + \alpha) \frac{2 + \varepsilon}{1 - \varepsilon} \int_0^t T_s^4(t') T_{m,0}^\alpha(t') G(R(t')) dt'} \times \left\{ \frac{3(4 + \alpha - \beta)x_{F,0}(t)\rho^{1+\lambda}}{8gT_{m,0}^\alpha(t)} \left[\frac{T_{\text{bath}}^4(t)}{T_s^4(t)} - 1 \right] - 1 + \frac{\varepsilon}{2} \right\}. \quad (4.5)$$

The bath drive therefore adds only one physical scalar ODE. It uses $T_{\text{bath}}(t)$ but not $\dot{T}_{\text{bath}}(t)$, which is advantageous for tabulated data. In Python, solve (4.5) together with (2.2) using `solve_ivp` with a stiff method. The cold onset is formally singular because the base-front history vanishes; a robust implementation starts at the first positive drive time with a small positive thermal seed and checks insensitivity to that seed.

After $T_s(t)$ and $B(t)$ have been determined, the surface- and bath-driven models are identical.

5 Monotone material-profile closure

The material profile is represented by the same radiation shape multiplied by one monotone ratio factor,

$$T(x, t) = T_{m,0}(t) \left[\left(1 - \frac{x}{x_F(t)} \right) \left(1 + B(t) \frac{x}{x_F(t)} \right) \right]^{1/(4+\alpha-\beta)} \times \exp \left[-s(t) \frac{x}{x_F(t)} \right]. \quad (5.1)$$

The ratio is

$$\frac{T(x, t)}{T_r(x, t)} = \frac{T_{m,0}(t)}{T_s(t)} \exp \left[-s(t) \frac{x}{x_F(t)} \right]. \quad (5.2)$$

Positive s narrows the material profile during heating, $s = 0$ gives proportional profiles, and negative s permits retained material energy during cooling. This parametrization remains positive for every real s and does not change the leading front exponent.

6 Profile moments and their closed forms

Every spatial integral required by the reduced model can be generated from two moments,

$$M_p(t, z) = \int_0^1 [(1-y)(1+B(t)y)]^{p/(4+\alpha-\beta)} e^{-zy} dy, \quad (6.1)$$

$$N_p(t, z) = \int_0^1 y [(1-y)(1+B(t)y)]^{p/(4+\alpha-\beta)} e^{-zy} dy. \quad (6.2)$$

These integrals have exact forms in terms of Humbert's confluent hypergeometric function Φ_1 :

$$M_p(t, z) = \frac{1}{1 + \frac{p}{4 + \alpha - \beta}} \Phi_1 \left(1, -\frac{p}{4 + \alpha - \beta}; 2 + \frac{p}{4 + \alpha - \beta}; -B(t), -z \right), \quad (6.3)$$

$$N_p(t, z) = \frac{1}{\left(1 + \frac{p}{4 + \alpha - \beta} \right) \left(2 + \frac{p}{4 + \alpha - \beta} \right)} \Phi_1 \left(2, -\frac{p}{4 + \alpha - \beta}; 3 + \frac{p}{4 + \alpha - \beta}; -B(t), -z \right). \quad (6.4)$$

The original integral representation of Humbert's function is

$$\Phi_1(A, B; C; x, z) = \frac{\Gamma(C)}{\Gamma(A)\Gamma(C-A)} \int_0^1 e^{zu} u^{A-1} (1-u)^{C-A-1} (1-xu)^{-B} du. \quad (6.5)$$

The conditions relevant here are $B(t) > -1$ and $p/(4 + \alpha - \beta) > -1$.

SciPy provides `hyp2f1` for $G(R)$ but does not currently expose Humbert Φ_1 directly. Although Φ_1 can be represented by nested series of one-variable hypergeometric functions, that is not the recommended production route. The most transparent and accurate Python evaluation of (6.1)–(6.2) is `quad`, which uses adaptive QUADPACK integration and returns an error estimate. The algebraic endpoint behavior can be supplied through `weight="alg"`. If performance later matters, a fixed Gauss–Jacobi rule can be validated against `quad`.

7 Areal material-energy equation

The material energy per unit transverse area represented by the reduced profile is

$$\mathcal{U}_m(t) = x_{F,0}(t) f \rho^{1-\mu} T_{m,0}^\beta(t) M_\beta(t, \beta s(t)). \quad (7.1)$$

The base front $x_{F,0}$ is used in this closure so that the material-energy subsystem remains sequential.

Integrating the material equation (1.4) over the reduced profile gives

$$\dot{\mathcal{U}}_m(t) = \frac{a_{\text{rad}} c}{g'} \rho^{1+\lambda'} x_{F,0}(t) \left[T_s^4(t) T_{m,0}^{-\alpha'}(t) M_{4-\alpha'}(t, -\alpha' s(t)) - T_{m,0}^{4-\alpha'}(t) M_{4-\alpha'}(t, (4-\alpha') s(t)) \right]. \quad (7.2)$$

At every ODE right-hand-side evaluation, $s(t)$ is determined from

$$M_\beta(t, \beta s(t)) = \frac{\mathcal{U}_m(t)}{x_{F,0}(t) f \rho^{1-\mu} T_{m,0}^\beta(t)}. \quad (7.3)$$

The left side is strictly decreasing because

$$\frac{\partial}{\partial s} M_\beta(t, \beta s) = -\beta \int_0^1 y [(1-y)(1+B(t)y)]^{\beta/(4+\alpha-\beta)} e^{-\beta s y} dy < 0. \quad (7.4)$$

Hence the root is unique. In Python, use the bracketed solver **brentq**. This is a local scalar inversion, not a Volterra equation; all material-energy history is compressed into the single ODE state $\mathcal{U}_m(t)$.

8 Final front correction

For any trial material-lag value s , define the first total-energy moment

$$\mathcal{E}_1(t, s) = a_{\text{rad}} T_s^4(t) N_4(t, 0) + f \rho^{1-\mu} T_{m,0}^\beta(t) N_\beta(t, \beta s). \quad (8.1)$$

Define also the dimensionless transport integrand

$$\mathcal{D}(t, s) = T_s^4(t) T_{m,0}^\alpha(t) \left\{ [1 - B(t)] M_\beta(t, \alpha s) + 2B(t) N_\beta(t, \alpha s) \right\}. \quad (8.2)$$

The compact form follows from the identity

$$\begin{aligned} & [(1-y)(1+By)]^{(\alpha+3)/(4+\alpha-\beta)} \left[-\frac{d}{dy} ((1-y)(1+By))^{1/(4+\alpha-\beta)} \right] \\ &= \frac{(1-B) + 2By}{4+\alpha-\beta} [(1-y)(1+By)]^{\beta/(4+\alpha-\beta)}. \end{aligned} \quad (8.3)$$

All dimensional constants in the physical transport moment are identical in the numerator and denominator below and therefore cancel.

The final moment-corrected front is

$$x_F^2(t) = x_{F,0}^2(t) \frac{\mathcal{E}_1(t, 0)}{\mathcal{E}_1(t, s(t))} \frac{\int_0^t \mathcal{D}(t', s(t')) dt'}{\int_0^t \mathcal{D}(t', 0) dt'}. \quad (8.4)$$

The denominator is simply the special case $s = 0$ of the same integrand. Numerically the two time integrals are two running scalar quadratures. Once $x_F(t)$ is known, the temperature profiles follow algebraically from (4.1) and (5.1).

9 Numerical implementation and complexity

9.1 Recommended SciPy functions

Table 1: Recommended Python functions for the reduced model.

Task	Function	Practical recommendation
Scalar ODE system	<code>solve_ivp</code>	Use <code>Radau</code> or <code>BDF</code> ; request dense output if profiles are needed at specified times.
Gray storage $G(R)$	<code>hyp2f1</code>	Evaluate the exact closed form (3.5); compare once against <code>quad</code> as a unit test.
Profile moments M_p, N_p	<code>quad</code>	Use tight <code>epsabs/epsrel</code> ; optionally use <code>weight="alg"</code> for the endpoint factor.
Scalar inversion for s	<code>brentq</code>	Bracket the unique root and reuse the previous s value to initialize the next bracket.
Tabulated drive interpolation	<code>PchipInterpolator</code>	Use the shape-preserving interpolant; for a prescribed T_s , obtain the needed derivative from <code>.derivative()</code> .

The current SciPy documentation for these functions is available at <https://docs.scipy.org/doc/scipy/reference/>.

9.2 Surface-temperature drive

For prescribed $T_s(t)$, the evolving quantities are

- the physical scalar ODE for $T_{m,0}(t)$;
- the physical scalar ODE for $\mathcal{U}_m(t)$;
- the running quadrature defining $x_{F,0}(t)$;
- the two running quadratures in (8.4).

At every ODE evaluation:

1. interpolate $T_s(t)$ and evaluate its derivative only for $B(t)$;
2. evaluate $R(t)$ and $G(R(t))$;
3. update the base-front quadrature and calculate $x_{F,0}(t)$;
4. calculate $B(t)$ from (4.2);
5. solve the unique scalar equation (7.3) for $s(t)$;
6. evaluate the moments and the right-hand sides for $T_{m,0}$, $\mathcal{U}_m$, and the three quadratures;
7. calculate $x_F(t)$ and the profiles only at requested output times.

Thus the surface-driven model requires

$$\boxed{2 \text{ scalar physical ODEs} + 3 \text{ running quadratures} + 1 \text{ scalar root.}} \quad (9.1)$$

9.3 Bath-temperature drive

For prescribed $T_{\text{bath}}(t)$, add only the scalar boundary ODE (4.5). The remaining calculation is unchanged. No derivative of the bath history is needed. The bath-driven model therefore requires

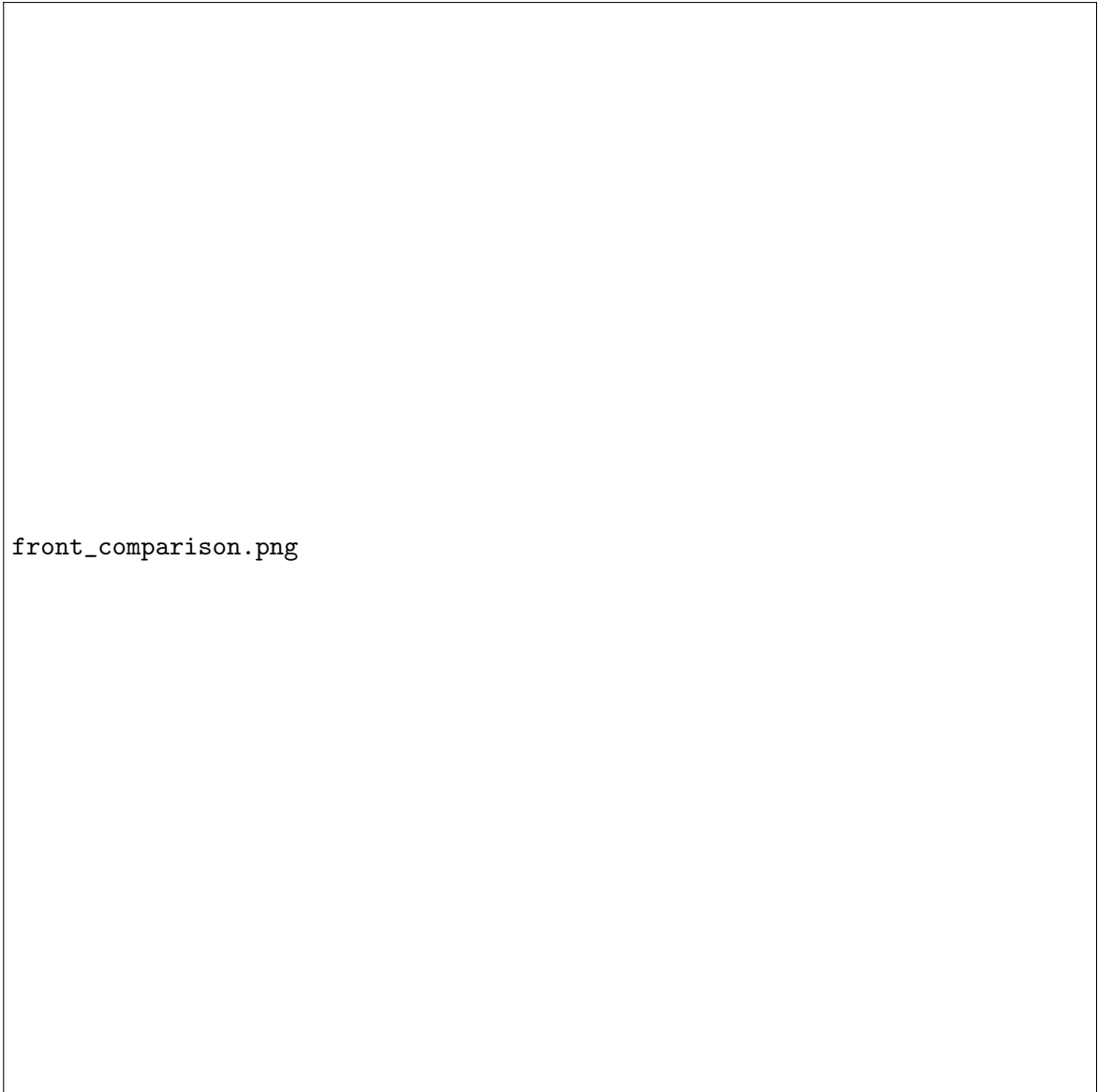
$$\boxed{3 \text{ scalar physical ODEs} + 3 \text{ running quadratures} + 1 \text{ scalar root.}} \quad (9.2)$$

The state dimension is fixed and independent of spatial resolution.

By comparison, a full gray PDE calculation requires hundreds or thousands of spatial cells, two unknown fields per cell, temperature-dependent face coefficients, an implicit matrix solve at every timestep, nonlinear opacity updates, boundary-face treatment, timestep control, and spatial convergence studies. The reduced model performs only a small fixed number of scalar operations per ODE evaluation.

10 Numerical illustration

The model was tested for the supplied SiO_2 drive using both surface and bath boundary specifications. The simulations use a validated $N = 1000$ finite-volume gray diffusion solver without a flux limiter. The bath-driven simulation uses the Marshak Robin boundary condition directly.



front_comparison.png

Figure 1: Surface- and bath-driven gray/LTE front comparisons through 4 ns. The unified gray-HR model uses the same interior closure for both gray drive types; only the boundary subsystem differs.

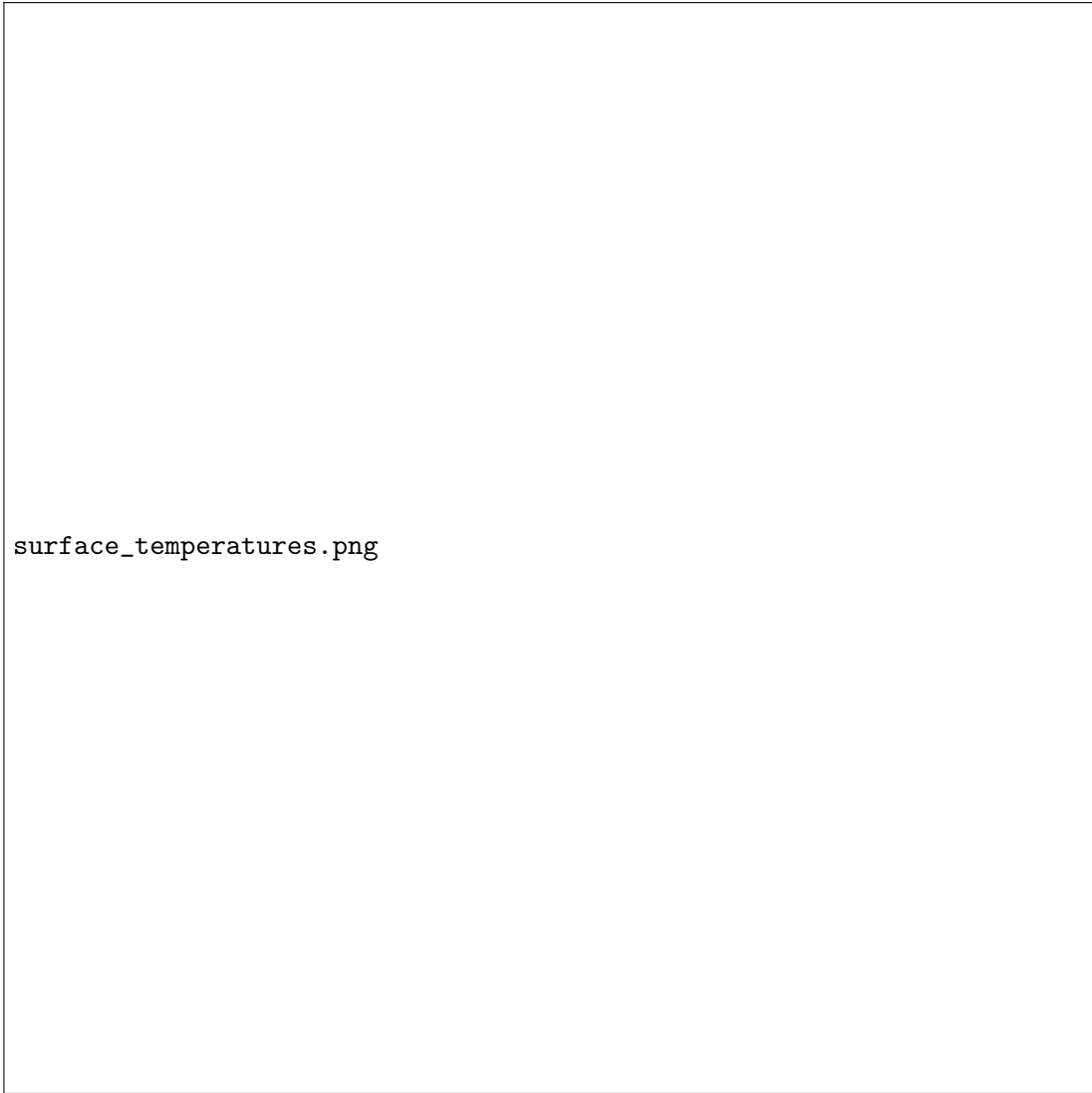
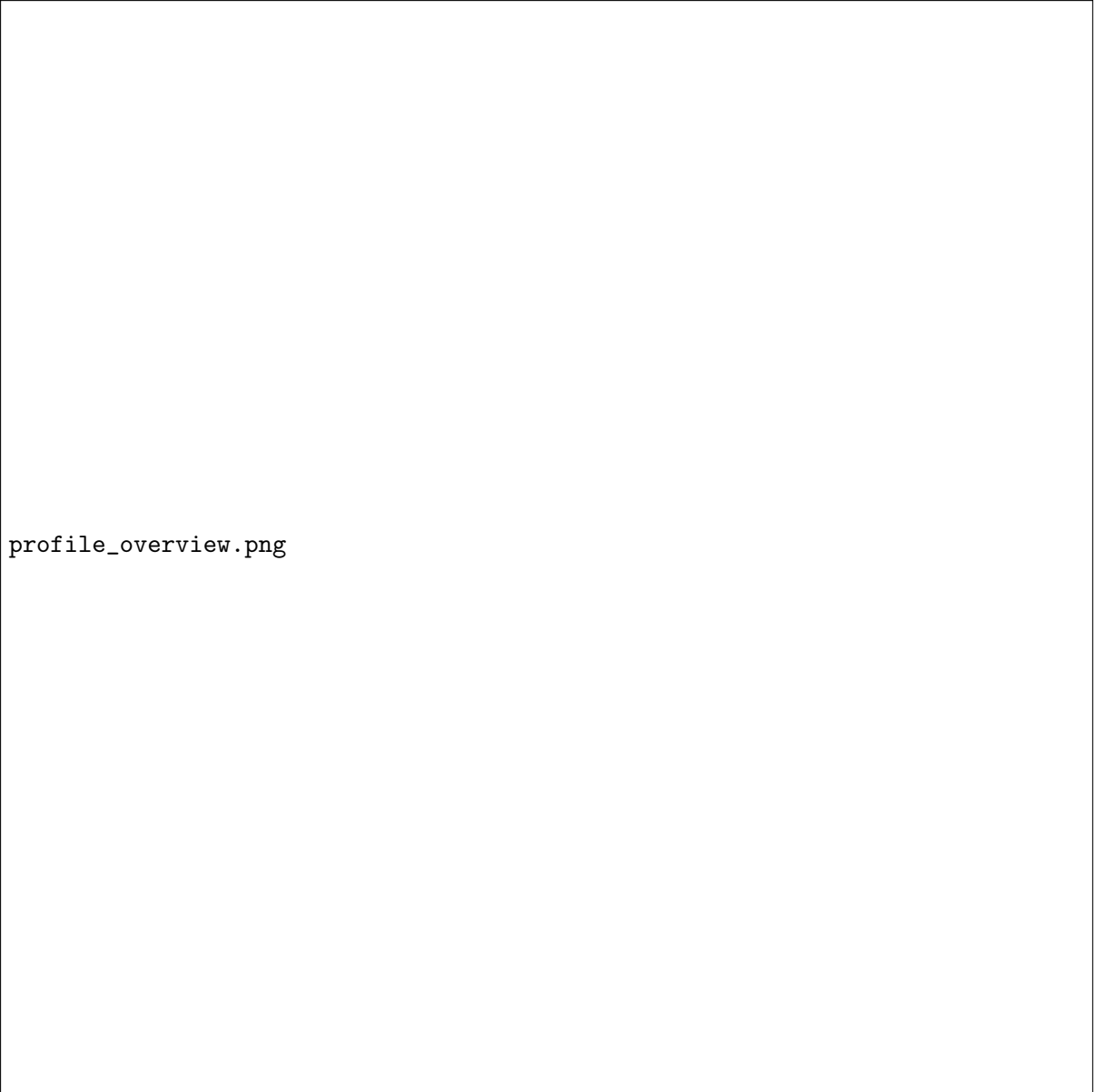


Figure 2: Prescribed bath temperature and the resulting surface radiation temperatures. The direct gray bath-HR boundary ODE follows the gray Robin simulation closely.



profile_overview.png

Figure 3: Gray radiation and material profiles for the surface-driven example. The monotone factor $\exp[-s(t)y]$ captures the early material narrowing without changing the front exponent.

For the bath-driven gray case, the front error decreases from approximately 8% at 0.5 ns to less than 1% near 4 ns. For the surface-driven gray case, the front remains near the one-percent level over most of the pulse. The larger relative errors immediately after the discontinuous turn-on occur inside the short onset layer where both the HR asymptotics and a numerical front definition are least reliable.

11 Accuracy and limitations

The model combines a first-order HR radiation profile with integrated material energy and transport moments. It is not a pointwise solution of the gray PDE and is not a proof of a uniform $\mathcal{O}(\varepsilon)$ expansion for arbitrary drives. Its principal approximations are:

- one HR-shaped radiation profile;
- one scalar material-lag parameter $s(t)$;
- use of the base front $x_{F,0}$ inside the surface-flux and material-energy closures;
- enforcement of integrated moments rather than the PDEs pointwise.

The model is expected to work best for sharp, single-front waves with smooth power-law material properties. Strong opacity transitions, multiple fronts, large excursions with $B(t) \leq -1$, or strongly non-HR radiation profiles indicate that another profile parameter or the full gray PDE may be required.

12 Self-contained summary of the working model

This section can be used independently as an implementation specification. No profile symbol is introduced; every profile and moment is written explicitly.

Problem definition and input

A fixed-density material occupies $x > 0$. Radiation and material temperatures are $T_r(x, t)$ and $T(x, t)$. Use

$$E_r = a_{\text{rad}} T_r^4, \quad U = f \rho^{1-\mu} T^\beta, \quad \boxed{a_{\text{rad}} = \frac{4\sigma_{\text{SB}}}{c}}, \quad (12.1)$$

$$\frac{1}{\kappa_R} = g T^\alpha \rho^{-\lambda}, \quad \frac{1}{\kappa_a} = g' T^{\alpha'} \rho^{-\lambda'}. \quad (12.2)$$

The Rosseland parameters are (g, α, λ) and the absorption parameters are (g', α', λ') . Define

$$\boxed{\varepsilon = \frac{\beta}{4 + \alpha}, \quad C = \frac{4a_{\text{rad}} c g}{3(4 + \alpha) f} \rho^{-2+\mu-\lambda}.} \quad (12.3)$$

Let

$$\boxed{T_{m,0}(t) = T(0, t), \quad T_s(t) = T_r(0, t).} \quad (12.4)$$

For a surface-temperature drive, $T_s(t)$ is prescribed. For a bath drive, $T_{\text{bath}}(t)$ is prescribed and $T_s(t)$ is determined by the Marshak condition

$$\boxed{F_s(t) = \frac{a_{\text{rad}} c}{2} [T_{\text{bath}}^4(t) - T_s^4(t)].} \quad (12.5)$$

Step 1: surface material temperature — scalar ODE

For either drive type, solve

$$\boxed{\frac{d}{dt} [f \rho^{1-\mu} T_{m,0}^\beta(t)] = \frac{a_{\text{rad}} c}{g'} \rho^{1+\lambda'} T_{m,0}^{-\alpha'}(t) [T_s^4(t) - T_{m,0}^4(t)].} \quad (12.6)$$

Numerically: use `solve_ivp(method="Radau")` or `solve_ivp(method="BDF")`.

Step 2: gray storage and base front — one running quadrature

Calculate

$$\boxed{R(t) = \frac{a_{\text{rad}} T_s^4(t)}{f \rho^{1-\mu} T_{m,0}^\beta(t)}.} \quad (12.7)$$

The gray storage factor may be evaluated either from its original integral,

$$\boxed{G(R) = (4 + \alpha - \beta) \int_0^1 \frac{\xi^{3+\alpha-\beta}}{1 + R \xi^{4-\beta}} d\xi,} \quad (12.8)$$

or from the exact closed form

$$G(R) = {}_2F_1\left(1, \frac{4+\alpha-\beta}{4-\beta}; 1 + \frac{4+\alpha-\beta}{4-\beta}; -R\right). \quad (12.9)$$

The defining integral of ${}_2F_1$ is

$${}_2F_1(a, b; c; z) = \frac{\Gamma(c)}{\Gamma(b)\Gamma(c-b)} \int_0^1 u^{b-1} (1-u)^{c-b-1} (1-zu)^{-a} du. \quad (12.10)$$

Numerically: use `hyp2f1`.

The base front is

$$x_{F,0}^2(t) = \frac{2+\varepsilon}{1-\varepsilon} C T_{m,0}^{-\beta}(t) \int_0^t T_s^4(t') T_{m,0}^\alpha(t') G(R(t')) dt'. \quad (12.11)$$

Numerically: store the time integral as one scalar quadrature state.

Step 3: radiation curvature $B(t)$

For prescribed $T_s(t)$,

$$B(t) = \frac{\varepsilon}{2} \left[1 - \frac{2+\varepsilon}{1-\varepsilon} \frac{\int_0^t T_s^4(t') T_{m,0}^\alpha(t') G(R(t')) dt'}{T_s^4(t) T_{m,0}^\alpha(t) G(R(t))} \times \frac{d}{dt} \ln T_s^{4+\alpha}(t) \right]. \quad (12.12)$$

Numerically: use a `PchipInterpolator` and its derivative for a tabulated $T_s(t)$.

For prescribed $T_{\text{bath}}(t)$, $T_s(t)$ is an additional scalar ODE,

$$\dot{T}_s(t) = \frac{2T_s^5(t) T_{m,0}^\alpha(t) G(R(t))}{\varepsilon(4+\alpha) \frac{2+\varepsilon}{1-\varepsilon} \int_0^t T_s^4(t') T_{m,0}^\alpha(t') G(R(t')) dt'} \times \left\{ \frac{3(4+\alpha-\beta)x_{F,0}(t)\rho^{1+\lambda}}{8gT_{m,0}^\alpha(t)} \left[\frac{T_{\text{bath}}^4(t)}{T_s^4(t)} - 1 \right] - 1 + \frac{\varepsilon}{2} \right\}. \quad (12.13)$$

and

$$B(t) = 1 - \frac{3(4+\alpha-\beta)x_{F,0}(t)\rho^{1+\lambda}}{8gT_{m,0}^\alpha(t)} \left[\frac{T_{\text{bath}}^4(t)}{T_s^4(t)} - 1 \right]. \quad (12.14)$$

Numerically: integrate $T_s(t)$ with the same stiff `solve_ivp` call. No derivative of $T_{\text{bath}}(t)$ is required.

Step 4: profile moments — adaptive scalar quadratures

For any p and z , define

$$M_p(t, z) = \int_0^1 [(1-y)(1+B(t)y)]^{p/(4+\alpha-\beta)} e^{-zy} dy, \quad (12.15)$$

$$N_p(t, z) = \int_0^1 y[(1-y)(1+B(t)y)]^{p/(4+\alpha-\beta)} e^{-zy} dy. \quad (12.16)$$

Their exact Humbert forms are

$$M_p(t, z) = \frac{1}{1 + \frac{p}{4+\alpha-\beta}} \Phi_1 \left(1, -\frac{p}{4+\alpha-\beta}; 2 + \frac{p}{4+\alpha-\beta}; -B(t), -z \right), \quad (12.17)$$

$$N_p(t, z) = \frac{1}{\left(1 + \frac{p}{4+\alpha-\beta}\right) \left(2 + \frac{p}{4+\alpha-\beta}\right)} \Phi_1 \left(2, -\frac{p}{4+\alpha-\beta}; 3 + \frac{p}{4+\alpha-\beta}; -B(t), -z \right), \quad (12.18)$$

with

$$\Phi_1(A, B; C; x, z) = \frac{\Gamma(C)}{\Gamma(A)\Gamma(C-A)} \int_0^1 e^{zu} u^{A-1} (1-u)^{C-A-1} (1-xu)^{-B} du. \quad (12.19)$$

Numerically: SciPy has no direct Φ_1 function, so use `quad` for M_p and N_p . The returned error estimate should be checked during validation.

Step 5: material lag $s(t)$ — one scalar ODE and one root

Evolve the areal material energy

$$\dot{\mathcal{U}}_m(t) = \frac{a_{\text{rad}} C}{g'} \rho^{1+\lambda'} x_{F,0}(t) \left[T_s^4(t) T_{m,0}^{-\alpha'}(t) M_{4-\alpha'}(t, -\alpha' s(t)) - T_{m,0}^{4-\alpha'}(t) M_{4-\alpha'}(t, (4-\alpha') s(t)) \right]. \quad (12.20)$$

At each ODE evaluation, solve

$$M_\beta(t, \beta s(t)) = \frac{\mathcal{U}_m(t)}{x_{F,0}(t) f \rho^{1-\mu} T_{m,0}^\beta(t)}. \quad (12.21)$$

Numerically: use `brentq`; the root is unique because the left side decreases strictly with s .

Step 6: final front — two running quadratures

For a trial value s , define

$$\mathcal{E}_1(t, s) = a_{\text{rad}} T_s^4(t) N_4(t, 0) + f \rho^{1-\mu} T_{m,0}^\beta(t) N_\beta(t, \beta s), \quad (12.22)$$

$$\mathcal{D}(t, s) = T_s^4(t) T_{m,0}^\alpha(t) \left\{ [1 - B(t)] M_\beta(t, \alpha s) + 2B(t) N_\beta(t, \alpha s) \right\}. \quad (12.23)$$

Then

$$x_F^2(t) = x_{F,0}^2(t) \frac{\mathcal{E}_1(t, 0)}{\mathcal{E}_1(t, s(t))} \frac{\int_0^t \mathcal{D}(t', s(t')) dt'}{\int_0^t \mathcal{D}(t', 0) dt'}. \quad (12.24)$$

Numerically: advance the numerator and denominator as two scalar running quadratures.

Step 7: final profiles — algebraic evaluation

For $0 \leq x \leq x_F(t)$,

$$T_r(x, t) = T_s(t) \left[\left(1 - \frac{x}{x_F(t)} \right) \left(1 + B(t) \frac{x}{x_F(t)} \right) \right]^{1/(4+\alpha-\beta)}, \quad (12.25)$$

$$T(x, t) = T_{m,0}(t) \left[\left(1 - \frac{x}{x_F(t)} \right) \left(1 + B(t) \frac{x}{x_F(t)} \right) \right]^{1/(4+\alpha-\beta)} \times \exp \left[-s(t) \frac{x}{x_F(t)} \right]. \quad (12.26)$$

Set both temperatures to zero beyond the sharp front.

Complete numerical workflow and cost

For prescribed $T_s(t)$:

1. integrate the two physical scalar ODEs for $T_{m,0}(t)$ and $\mathcal{U}_m(t)$;
2. advance the base-front quadrature and the two final-front quadratures;
3. evaluate G with `hyp2f1`;
4. evaluate M_p, N_p with `quad`;
5. solve the unique $s(t)$ root with `brentq`;
6. evaluate x_F, T_r , and T algebraically at output times.

The cost is

$$\boxed{2 \text{ scalar physical ODEs} + 3 \text{ scalar quadratures} + 1 \text{ scalar root.}} \quad (12.27)$$

For prescribed $T_{\text{bath}}(t)$, add only the scalar ODE for $T_s(t)$; all later steps are identical. The cost is

$$\boxed{3 \text{ scalar physical ODEs} + 3 \text{ scalar quadratures} + 1 \text{ scalar root.}} \quad (12.28)$$

No spatial grid, diffusion matrix, PDE iteration, or time-dependent integral equation appears in either reduced model.